PHY · Circular motion in horizontal and vertical planes — syllabus points and past-paper evidence

Every award of the 2016–2025 papers, labelled to the 2026/2027 Year 12 syllabus content (identical in both). Statistics from the labels; notes written by AI from that evidence: review before use.

Pattern 1: Horizontal, banked and vertical circular motion

Syllabus: S26-U3-SU-05 / S27-U3-SU-05

Frequency: 10 of 10 exam years; 119 marks in all.

Marks reward: Substitution and working 35, Select the principle or relationship 34, Qualitative explanation 19, Final answer 17, Diagram 10, Recall 3, Uncertainty and error 1

Past questions: 2016P-19(a); 2016P-19(b); 2016P-19(c); 2016R-13; 2017P-20(a); 2017P-20(b); 2017P-20(c); 2018C-18(a); 2018C-18(c); 2018R-9(a); 2018R-9(b); 2019P-16(a); 2019P-16(b); 2019P-16(c); 2019R-8; 2020C-20(dii); 2020C-21(c); 2021P-14(cii); 2021P-14(ciii); 2021P-17(a); 2021P-17(b); 2021P-17(c); 2021P-17(d); 2021R-9…

Summary: This point examines circular motion where a net force points towards the centre: uniform motion on a flat bend, banked tracks and flight turns, and vertical loops or pendulum-type paths, including apparent weight. It appears often as a multi-part calculation question that combines a free body diagram with a centripetal force equation, and it is frequently paired with a short explanation of why a force or speed changes.

Recognise: Look for an object moving along a curved path with a fixed radius, a string, track, bank or rotating frame, or a question asking for speed, period, frequency, tension, reaction force or apparent weight. Key words include centripetal, net force towards the centre, banked, conical, loop, top and bottom of the circle, and apparent weight.

Steps

1. Identify the centre of the circle and the radius r of the path. The radius of the path is not always the string length or track width; for a conical pendulum use r = L sin(theta).
2. Draw a free body diagram with only real forces (weight, tension, normal force, friction, upthrust). Do not draw a centripetal force as a separate arrow.
3. Resolve the forces into components towards the centre and vertically. The net inward force must equal the centripetal requirement F = mv^2/r, with v = 2pir/T and a = v^2/r.
4. For a banked surface with no vertical acceleration, balance the vertical forces, then use the horizontal equation: tan(theta) = v^2/(r*g) for the ideal case with no friction.
5. For vertical circular motion, apply the radial equation at the specific point: at the top with the object on the inside, the minimum speed occurs when contact force is zero, giving v = sqrt(r*g); at the bottom the tension or normal force must supply the weight plus mv^2/r.
6. Apparent weight equals the normal or contact reaction R from the surface or string at that point, not the true weight, and it changes with speed and position on the circle.
7. Check units before substituting (convert km/h to m/s, frequency to period with T = 1/f) and check the answer is sensible, for example that a larger radius requires less centripetal force for the same speed.

Where the marks are

- Marks for a labelled free body diagram with the correct number of real forces and no extra forces.
- Marks for writing the net inward force as the centripetal force equation, and resolving forces into radial and vertical components with the vertical balance shown for banking or conical cases.
- Marks for correct use of radius of path (not string length) and correct unit conversion.
- Marks for a clear explanation of why speed, angle or contact force changes, linked to the relevant equation.
- Marks for interpreting apparent weight as the contact force at the point asked, with direction or sign reasoning at top or bottom.

Pitfalls

- Drawing a centripetal force arrow on the free body diagram in addition to the real forces.
- Using the string length or track width as the radius of the circular path, especially for a conical pendulum or banked surface.
- Forgetting that the vertical forces must balance in a banked or conical case, or omitting the vertical component of tension or normal force.
- Mixing up the signs at the top and bottom of a vertical loop, for example writing tension as mv^2/r - mg at the bottom.
- Failing to convert km/h to m/s or confusing frequency with period when substituting into v = 2pir/T.

Pattern 2: Energy conservation in circular motion

Syllabus: S26-U3-SU-06 / S27-U3-SU-06

Frequency: 6 of 10 exam years; 21 marks in all.

Marks reward: Substitution and working 9, Final answer 7, Select the principle or relationship 4, Qualitative explanation 1

Past questions: 2016C-22(cii); 2018P-14(a); 2019P-12(a); 2019P-16(d); 2020C-21(b); 2022P-13(d); 2022P-14(c); 2022R-1; 2025P-17(c)

Summary: This point tests applying conservation of energy (or the work-energy relation W = ΔE when non-conservative forces act) to find the speed of an object at a different point on a circular or swinging path. It usually appears as a multi-step calculation: find the height change from geometry, then solve for speed, and sometimes as a short written explanation of the energy transfers.

Recognise: Look for a mass moving on a string, track, or pendulum-type arc, with a radius or length and an angle or two heights given. The question asks for a speed at a lower or higher point, or at the bottom after release. Wording such as 'using conservation of energy' or 'assume no energy is lost' is a strong signal. Questions asking whether a string or track will stay taut at a point combine energy with the circular-motion condition.

Steps

1. Identify the two points involved (for example release and the point of interest) and choose a reference level for gravitational potential energy, usually the lowest point of the path.
2. Find the change in height Δh between the two points from geometry. For a string of length L released from angle θ to the vertical, $\Delta h = L(1 - \cos\theta)$. Use the vertical drop, not the path length or string length.
3. Write the energy balance: $\frac{1}{2}mv_1^2 + mgh_1 = \frac{1}{2}mv_2^2 + mgh_2$. If only gravity does work, this gives $v_2^2 = v_1^2 + 2g\Delta h$, where mass cancels.
4. If friction or another non-conservative force does work, use W = ΔE instead, with the work done by friction as negative work ($W = -F \cdot d$) and the energy change of the system set equal to it.
5. Rearrange algebraically for v before substituting numbers, and take the positive square root for speed.
6. Convert all quantities to SI units first (kg, m, J, m s^{-1}), especially energies given in eV or keV, then substitute and round only at the end.
7. Check the answer is sensible: speed should be larger after descending and smaller after rising.

Where the marks are

- A mark for stating the energy conservation equation or the W = ΔE relation before substituting numbers.
- A mark for a correct height change obtained from geometry, shown explicitly.
- A mark for rearranging the equation to isolate v, often separate from the substitution mark.
- A mark for the final speed with correct units and sensible significant figures.
- When asked to explain, a mark for stating in words that gravitational potential energy converts to kinetic energy (or that input energy equals output energy).

Pitfalls

- Using the string length or arc length as the height change instead of the vertical drop $L(1 - \cos\theta)$.
- Dropping the ½ in kinetic energy, or giving v^2 as the final answer without taking the square root.
- Applying energy conservation when friction or another non-conservative force does work, and ignoring the energy lost.
- Mixing reference levels, so the potential energy at one point is measured from a different level than the other.
- Failing to convert units such as eV, keV or grams to SI before calculating, which gives wrong energies by large factors.